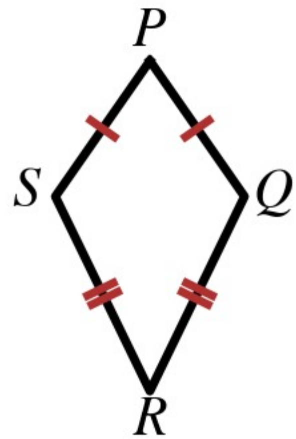
Name: ____________________________________

Kite PQRS at the right: $\overline{PQ} \cong \overline{PS}$ and $\overline{RQ} \cong \overline{RS}$.



1. Which pair of opposite angles must be congruent?

2. $m\angle P = 62°$ and $m\angle Q = 113°$. Find $m\angle R$.

3. What is true of the diagonals of every kite?

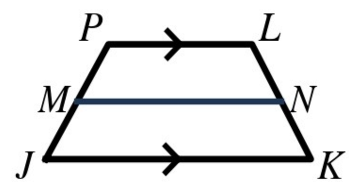
Trapezoids.

4. In trapezoid ABCD, $\overline{AB} \parallel \overline{DC}$ and $m\angle A = 74°$. Find $m\angle D$.

5. Trapezoid EFGH is isosceles with base angles $\angle E$ and $\angle F$. $m\angle E = (7x - 4)°$ and $m\angle F = (5x + 18)°$. Find x and $m\angle E$.

6. The legs of a trapezoid measure $9y - 14$ and $6y + 13$. Find y so the trapezoid is isosceles.

7. An isosceles trapezoid has diagonals $4z + 9$ and $7z - 15$. Find z and the length of each diagonal.



Midsegment $\overline{MN}$ of trapezoid JKLP at the right.

8. The bases measure 34 and 22. Find MN.

9. MN = 26 and one base measures 19. Find the other base.

10. The bases measure $11x - 8$ and $7x + 2$, and MN = 42. Find x.